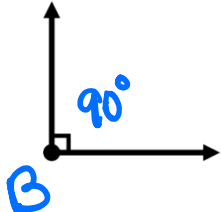
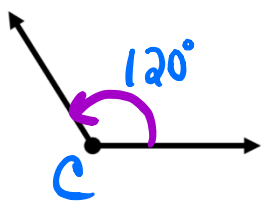
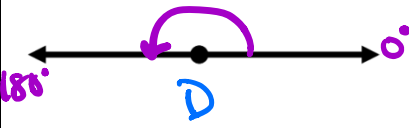
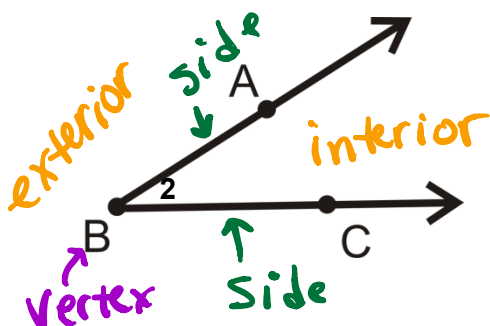


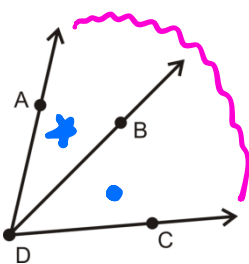
Unit 1 - Section 1.5 - Measuring and Constructing Angles

Types of Angles

<u>Picture</u> (use a protractor to measure)	<u>Type and Definition</u>
	<u>Acute</u> $0^\circ < \text{angle measure} < 90^\circ$
	<u>Right</u> $m\angle B = 90^\circ$ Always = 90°
	<u>Obtuse</u> $90^\circ < \text{angle measure} < 180^\circ$
	<u>Straight</u> Always 180°

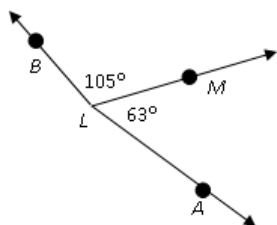
AnglesSides
 $\overrightarrow{BA}$
 $\overrightarrow{BC}$
Naming
 $\angle B$
 $\angle ABC$
 $\angle CBA$
 $\angle 2$

Angle Addition Postulate



$$m\angle ADB + m\angle BDC = m\angle ADC$$

Example 1: Find the measure of $\angle BLA$.

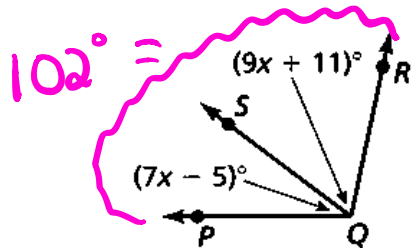


$$m\angle BLM + m\angle MLA = m\angle BLA$$

$$105^\circ + 63^\circ = 168^\circ$$

$$m\angle BLA = 168^\circ$$

Example 2: Given that $\angle PQR = 102^\circ$, find $m\angle SQR$ and $m\angle PQS$.



$$(7x - 5) + (9x + 11) = 102$$

$$16x + 6 = 102$$

$$7(6) - 5$$



$$9(6) + 11$$



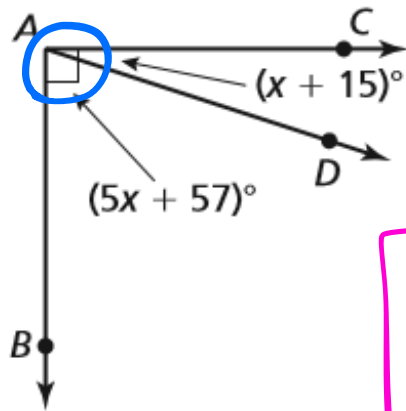
$$x = 6$$

$$m\angle PQS = 37^\circ \quad m\angle SQR = 65^\circ$$

Example 3:

Find $m\angle CAD$
and $m\angle BAD$.

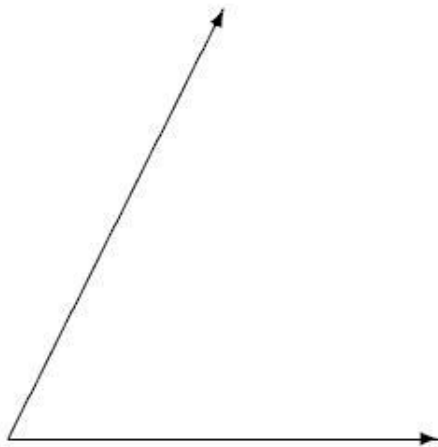
$$5x + 57 + x + 15 = 90$$



$$x = 3$$

$$\begin{aligned} m\angle CAD &= 18^\circ \\ m\angle BAD &= 72^\circ \end{aligned}$$

Use a Compass and Straightedge



1. Copy the Angle (<http://mathopenref.com/constcopyangle.html>)
2. Construct an Angle Bisector
(<http://www.mathopenref.com/constbisectangle.html>)
3. Measure all angles.

